



(Year 2 / A2 Level)

Nuclear Physics Lecture Notes

edited by Learmates

originally made by *Yuhao Yang*

CHAPTER 23

Nuclear Physics

23.1 nuclear energy

23.1.1 mass-energy equivalence

mass-energy equivalence principle states that mass and energy are equivalent to one another

this idea is represented mathematically by **Einstein's mass-energy relation**: $E = mc^2$ ^[1]

m is **rest mass** of the object, $c = 3.00 \times 10^8 \text{ m s}^{-1}$ is speed of light in vacuum

➤ mass-energy equivalence implies mass can be converted into pure energy

conversely, mass can be created out of energy

➤ $\Delta E = \Delta mc^2$ applies to *all* energy changes

if energy is supplied to a system, then mass of this system increases

for example, an object will have a greater mass when it is set to motion or is heated

mass is not a conserved quantity, it is *mass-energy* that is conserved in all processes

➤ if an object is stationary, its mass is called the *rest mass* m_0

mass-energy equivalence implies an object at rest still has an intrinsic *rest energy*: $E_0 = m_0 c^2$

➤ energy transformations such as chemical reactions can cause a system to lose some mass content, but this change in mass is usually negligible

^[1]This might be the most famous equation in all physical sciences. When Albert Einstein wrote down the theory of *special relativity* in 1905, he found the mass m of an object with rest mass m_0 is related to the speed v at which it is moving by: $m = m_0 \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}}$. Total energy is given by $E = mc^2 = m_0 c^2 \left(1 - \frac{v^2}{c^2}\right)^{-\frac{1}{2}}$.

At low speeds $v \ll c$, $E \approx m_0 c^2 \left(1 + \frac{1}{2} \frac{v^2}{c^2}\right) = m_0 c^2 + \frac{1}{2} m_0 v^2$. The second piece clearly gives the kinetic energy term, so the first piece can be thought as the rest energy stored within the mass. This is how the idea of mass-energy relation came to the great mind of Einstein.

Example 23.1 When an antiproton (antiparticle of a proton) collides with a proton, they are annihilated and two photons of equal energy are formed. (a) What is the energy of each photon? (b) What is the photon frequency?

 rest energy of proton becomes photon energy:

$$E_\gamma = m_p c^2 = 1.67 \times 10^{-27} \times (3.0 \times 10^8)^2 \approx 1.50 \times 10^{-10} \text{ J}$$

$$\text{photon frequency: } f = \frac{E_\gamma}{h} = \frac{1.50 \times 10^{-10}}{6.63 \times 10^{-34}} \approx 2.27 \times 10^{23} \text{ Hz (high-frequency } \gamma\text{-photon)} \quad \square$$

Question 23.1 The combustion of one mole of solid carbon to form carbon dioxide (CO_2) at standard condition is 394 kJ. Find the change in mass for this amount of energy, and hence compare this mass change with the mass of carbon before combustion.

Question 23.2 Given that the specific heat capacity of copper is $380 \text{ J kg}^{-1} \text{ K}^{-1}$. When a copper block is heated from 300 K to 1000 K, what is the additional mass as a fraction of its rest mass?

23.1.2 binding energy

nucleons (protons and neutrons) in a nucleus bind together through the *strong nuclear force* to pull nucleons apart, work must be done to overcome the attraction so free nucleons at infinity have greater potential energy than a single nucleus since energy is equivalent to mass, free nucleons would appear more massive than when they are held together in a nucleus

this statement is supported by experimental data

useful notions in nuclear physics related to this idea can now be introduced

difference between mass of a nucleus and total mass of its constituent nucleons when separated to infinity is called the **mass defect**

energy needed to separate the nucleons in a nucleus to infinity, or equivalently, energy released when free individual nucleons combine to form a nucleus, is called the **nuclear binding energy** (E_B)

➤ for a nucleus Z with proton number Z and nucleon number A (represented by A_ZX)

its mass defect is given by: $\Delta m = Zm_p + (A - Z)m_n - m_X$

where $m_p = 1.673 \times 10^{-27}$ kg is mass of proton, $m_n = 1.675 \times 10^{-27}$ kg is mass of neutron

➤ by definition, binding energy is equivalent to mass defect: $E_B = \Delta mc^2$

➤ nucleons in a nucleus have *negative* P.E., while free nucleons have zero P.E.

E_B is the energy required to fill this gap in order to pull nucleons apart

i.e., E_B of a nucleus equals *loss* of potential energy during its formation

➤ nuclear mass is often measured in *unified atomic mass* u, where $1 \text{ u} = 1.66 \times 10^{-27} \text{ kg}$

nuclide ${}_Z^AX$ has a nuclear mass of about Au

➤ binding energy is often measured in MeV, where $1 \text{ MeV} = 10^6 \text{ eV} = 1.60 \times 10^{-13} \text{ J}$

Example 23.2 An iron-56 nucleus (${}_{26}^{56}\text{Fe}$) has a mass of 9.288×10^{-26} kg. Calculate the nuclear binding energy per nucleon, in MeV, for ${}_{26}^{56}\text{Fe}$.

 mass defect: $\Delta m = 26m_p + (56 - 26)m_n - m_{\text{Fe}} = 26 \times 1.673 \times 10^{-27} + 30 \times 1.675 \times 10^{-27} - 9.288 \times 10^{-26}$

so we find $\Delta m = 8.68 \times 10^{-28}$ kg

binding energy: $E_B = \Delta mc^2 = 8.68 \times 10^{-28} \times (3.00 \times 10^8)^2 \approx 7.812 \times 10^{-11} \text{ J}$

binding energy per nucleon: $\epsilon_b = \frac{E_b}{A} = \frac{7.812 \times 10^{-11}}{56} \approx 1.395 \times 10^{-12} \text{ J}$

convert into MeV: $\epsilon_b = \frac{1.395 \times 10^{-12}}{1.60 \times 10^{-13}} \text{ MeV} \approx 8.72 \text{ MeV}$ □

Question 23.3 Show that the energy equivalent of 1.0 u is 934 MeV.

Question 23.4 Given that mass of proton is 1.007 u, mass of neutron is 1.009 u, and mass of uranium-235 nucleus is 234.992 u. Find the binding energy per nucleon of nuclide ${}_{92}^{235}\text{U}$.

Question 23.5 What is the binding energy for the hydrogen nucleus ${}_1^1\text{H}$?

Question 23.6 Why is rest mass of proton slightly larger than the unified atomic mass unit?

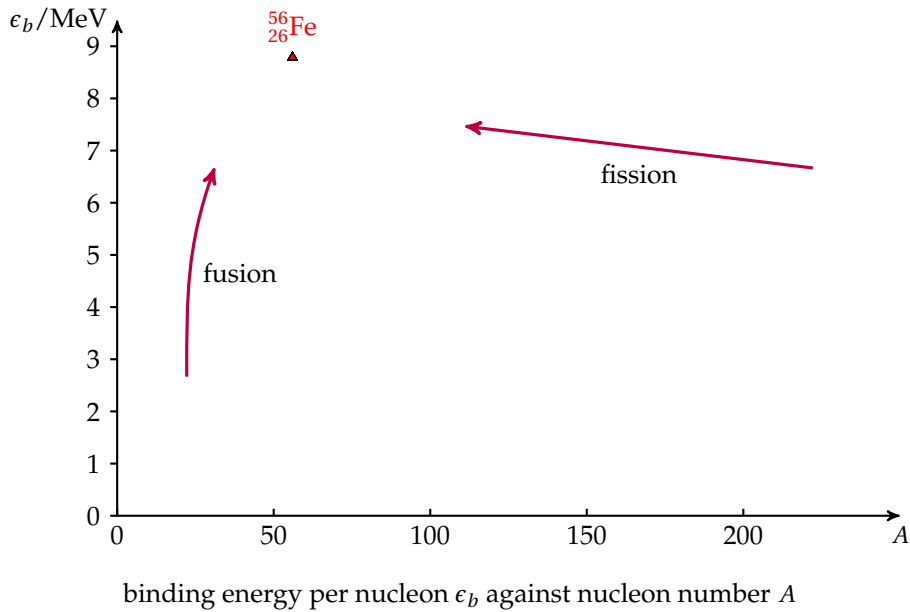
23.1.3 nuclear stability

binding energy per nucleon ϵ_b is closely related to nuclear stability

binding energy per nucleon gives average energy needed to remove a nucleon from nucleus

higher ϵ_b means more difficult to pull nucleons away, so nucleus has higher stability

a graph of ϵ_b against nucleon number A can be plotted based on experimental data



- ${}^{56}_{26}\text{Fe}$ has greatest value of ϵ_b for all nucleus $\Rightarrow {}^{56}_{26}\text{Fe}$ is the most stable nuclide in nature
- any physical system would evolve such that it lowers total energy
if there could be an increase in total binding energy, nuclear reactions could occur

23.1.4 fusion

for nuclei smaller than ${}^{56}_{26}\text{Fe}$, they could combine together and release energy

fusion is the process where two light nuclei join together to form a heavier nucleus

- fusion occurs only at very *high temperatures*
nuclei are all positively charged, for two nuclei to come close enough to fuse together, high initial K.E. is needed for them to overcome their *electrostatic repulsion*
- vast amount of energy of the sun and other *stars* are created through fusion
extreme high temperature and high density at core of stars make fusion possible ^[2]

^[2] Even for the simplest fusion reaction, fusing hydrogen into helium, at least 1.5×10^7 K is required. This is basically how the sun shines and nurtures lives on the earth, and also many other stars in the universe power energies. For a sufficiently massive star, when it burns out of its fuel, its core collapses and becomes hot enough to start to fuse heavier elements. If a star is not able to fuse again, nuclear reaction ceases, the star collapses and becomes a *white dwarf*. For a giant star, nuclear reactions can go all the way up to ${}^{56}_{26}\text{Fe}$. At this point, no more energy can be produced through fusion. The star collapses extremely rapidly and then explodes, creating a *supernova*, during which all elements heavier than ${}^{56}_{26}\text{Fe}$

23.1.5 fission

for a nucleus larger than ${}^{56}_{26}\text{Fe}$, it can break into two or more parts and release energy

fission is the process where a massive nucleus splits into two smaller nuclei of about the same size (and is usually associated with release of several neutrons)

- note that fission requires the two product nuclei are of similar size
 - α -decay (emission of helium nucleus from an unstable nucleus) is not a fission process
- fission reactions can occur *spontaneously*, i.e. nuclei can undergo fission by themselves
 - fission can also be *induced* by bombarding the nucleus with an incident neutron
 - induced fission is used by humans to generate nuclear power or to build nuclear weapons

nuclear reactors (*)

nuclear reactors make use of energy released from induced fission reactions

thermal energy produced from fission is further converted to electrical or mechanical forms

uranium-235 (${}^{235}_{92}\text{U}$) is the most widely used fuel for fuel nuclear reactors

one of the many fission reactions of uranium-235 is: ${}^{235}_{92}\text{U} + {}^1_0\text{n} \longrightarrow {}^{141}_{56}\text{Ba} + {}^{92}_{36}\text{Kr} + 3{}^1_0\text{n}$

- each reaction releases more than one neutrons
 - they can trigger further fission reactions, making *chain reaction* possible
 - hence huge amount of energy can be released in a very short time
- rate of fission should be controlled, a runaway reaction could lead to disastrous explosion
 - control rods* (e.g., boron) are used to absorb neutrons: ${}^{10}_5\text{B} + {}^1_0\text{n} \longrightarrow {}^7_3\text{Li} + {}^4_2\text{He}$
 - control rods are adjusted so that one neutron per reaction goes on to produce further fission
 - in emergency, release of boron rods shuts down reactor
- reaction is expected to continue at steady rate
 - need sufficient amount of neutrons to maintain the chain reaction
 - this requires a *critical mass* for the amount of ${}^{235}_{92}\text{U}$ fuel

are formed. (*)

- only low energy neutrons (*thermal neutrons*) can be *captured* by $^{235}_{92}\text{U}$
 - but neutrons released through fission are very energetic
 - a *moderator* (e.g., water) is needed to slow down neutrons
- heat produced in fission is removed by *coolant* (e.g., water)
 - this heat can be used to power generators to produce electricity
- reactor is surrounded by a *shield* (e.g., a thick concrete) to prevent radiation from escaping

23.1.6 energy release from nuclear reactions

in this section, we consider only those nuclear reactions that release energy

let's write a generic nuclear reaction as: original particles $\longrightarrow$ product particles + Q

two approaches to compute Q , the amount of energy release, will be introduced

method 1: use change in *mass* during the reaction

there must be a decrease in total mass to be converted into energy release

using mass-energy relation: $Q = \Delta mc^2 = (m_{\text{org}} - m_{\text{prod}})c^2$

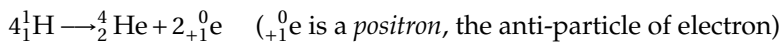
method 2: use change of *total binding energy*

energy is released means product particles are more stable

so total binding energy of product particles is higher than that of original particles

energy released during reaction: $Q = E_{B,\text{prod}} - E_{B,\text{org}}$

Example 23.3 The sun is a huge nuclear reactor. It fuses hydrogen nuclei into helium nuclei to produce large amounts of energy. The reaction can be expressed by the formula:



(a) Evaluate the energy released in one reaction. (b) Given that the power output of the sun is roughly $4.0 \times 10^{26} \text{ W}$, estimate how much hydrogen the sun burns every second. (Data for this question: $m_{\text{p}} = 1.672622 \times 10^{-27} \text{ kg}$, $m_{\text{He}} = 6.644657 \times 10^{-27} \text{ kg}$, $m_{\text{e}} = 9.11 \times 10^{-31} \text{ kg}$)

 energy released in one reaction:

$$E_0 = \Delta mc^2 = (4m_{\text{H}} - m_{\text{He}} - 2m_{\text{e}})c^2 \approx 4.40 \times 10^{-29} \times (3.0 \times 10^8)^2 \approx 3.96 \times 10^{-12} \text{ J}$$

in interval Δt , sun outputs a total energy of $P\Delta t$ via a number of ΔN fusion reactions

we write $P\Delta t = \Delta NE_0$, so number of fusion reactions every second is:

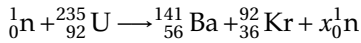
$$\frac{\Delta N}{\Delta t} = \frac{P}{E_0} = \frac{4.0 \times 10^{26}}{3.96 \times 10^{-12}} \approx 1.0 \times 10^{38} \text{ s}^{-1}$$

each reaction takes four hydrogen nuclei, so rate of hydrogen consumption:

$$\frac{\Delta M_{\text{H}}}{\Delta t} = \frac{4\Delta N}{\Delta t} \times m_{\text{H}} \approx 4 \times 1.0 \times 10^{38} \times 1.67 \times 10^{-27} \approx 6.7 \times 10^{11} \text{ kg s}^{-1}$$

the sun burns over 600 billion kilograms of hydrogen every second, just think about it! $\square$

Example 23.4 Uranium-235 nuclei ($^{235}_{92}\text{U}$) bombarded with slow neutrons can undergo nuclear reaction:



(a) Determine the number of x . (b) Using the data in

	binding energy per nucleon
$^{235}_{92}\text{U}$	7.591 MeV
$^{141}_{56}\text{Ba}$	8.326 MeV
$^{92}_{36}\text{Kr}$	8.513 MeV

the table, find the energy released for one reaction. (c) What is the change in mass, if any, before and after the reaction?

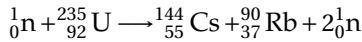
 from conservation of mass number: $1 + 235 = 141 + 92 + x \Rightarrow x = 3$

energy release during reaction equals change in total binding energy:

$$Q = E_{B,\text{Ba}} + E_{B,\text{Kr}} - E_{B,\text{U}} = 141 \times 8.326 + 92 \times 8.513 - 235 \times 7.591 = 173.277 \text{ MeV}$$

$$\text{reduction in total mass: } \Delta m = \frac{Q}{c^2} = \frac{173.277 \times 1.60 \times 10^{-13}}{(3.0 \times 10^8)^2} \approx 3.08 \times 10^{-28} \text{ kg} \quad \square$$

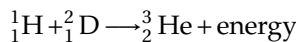
Question 23.7 Uranium-235 nuclei can undergo another fission process when being bombarded with neutrons. This is represented by:



Calculate the energy released in this reaction.

	binding energy per nucleon
$^{235}_{92}\text{U}$	7.591 MeV
$^{144}_{55}\text{Cs}$	8.212 MeV
$^{92}_{36}\text{Kr}$	8.631 MeV

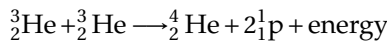
Question 23.8 The *proton-proton chain reaction* is a set of nuclear reactions through which stars fuse hydrogen into helium. One intermediate reaction may be summarised as:



Calculate the energy released in this reaction.

	mass/u
^1_1H	1.00728
^2_1D	2.01356
^3_2He	3.01605

Question 23.9 Another fusion reaction in the proton-proton chain is represented by:



where energy released through this reaction is 12.86 MeV. Given that binding energy per nucleon of ${}^4_2\text{He}$ is 7.074 MeV, calculate the binding energy per nucleon for ${}^3_2\text{He}$.

Question 23.10 If the total mass of the product particles is greater than that of the original particles in a nuclear reaction, what can you say about the energy of the original particles?

23.2 radioactive decay

23.2.1 radioactive decay

the process of *random* and *spontaneous* emission of α -, β -, and γ -radiation from unstable nuclei is called **radioactive decay**

➤ **random** means decay events are not predictable

we cannot tell precisely when a particular nucleus is about to decay

we can only tell the *probability* of decay within a certain time

➤ **spontaneous** means rate of decay does not depend on external conditions

temperature, pressure, or has no effect on rate of radioactive decays

suppose initially a sample contains N radioactive nuclei that are about to decay

after a time interval Δt , ΔN nuclei undergo decay processes

probability of decay for nuclei in the sample during this time interval is $\frac{\Delta N}{N}$

divide by Δt , probability of decay for each nucleus per unit time can be found

since nuclear decay is spontaneous, this number is a constant, called the decay constant

decay constant is the probability for one nucleus to decay per unit time: $\lambda = \frac{\Delta N}{N\Delta t}$

➤ to describe the rate of decay, we introduce the notion of **activity**

activity is defined as the number of nuclei that undergo decay per unit time: $A = \frac{\Delta N}{\Delta t}$

this can be given in a differential form: $A = -\frac{dN}{dt}$

a minus sign is included because number of nuclei N decreases with time

➤ compare with the expression for decay constant, we find: $A = \lambda N$

this equation shows that if there are more nuclei present, the sample has greater activity

➤ units for decay constant and activity: $[\lambda] = [A] = \text{s}^{-1}$

for activity, one decay event per unit time is defined as one *becquerel*: $1 \text{ Bq} = 1 \text{ s}^{-1}$

23.2.2 decay equations

variation of number of undecayed nuclei with time can be derived from the equation: $A = \lambda N$

recall that $A = -\frac{dN}{dt}$, so this equation is a differential equation in disguise: $-\frac{dN}{dt} = \lambda N$

this can be solved by separating variables and integrating

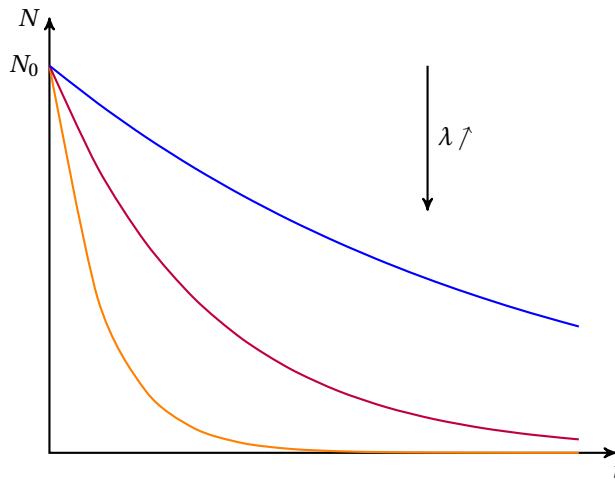
$$-\lambda dt = \frac{dN}{N} \Rightarrow -\lambda \int_0^t dt = \int_{N_0}^N \frac{dN}{N} \Rightarrow -\lambda t \Big|_0^t = \ln N \Big|_{N_0}^N \Rightarrow -\lambda t = \ln \frac{N}{N_0} \Rightarrow N = N_0 e^{-\lambda t}$$

this shows that decay events obey *exponential decay laws*

➤ number of undecayed nuclei at time t is given by: $N(t) = N_0 e^{-\lambda t}$

N_0 is initial number of nuclei in the sample at $t = 0$

➤ $\lambda \uparrow \Rightarrow$ higher probability of decay, or greater rate of decay



exponential decay law for number of nuclei: $N(t) = N_0 e^{-\lambda t}$

➤ since activity $A = \lambda N$, so activity varies with time as: $A(t) = A_0 e^{-\lambda t}$

Example 23.5 A sample containing 6.0×10^7 iodine-131 nuclei has an activity of 60 Bq. (a) What is the decay constant of iodine-131? (b) How many nuclei remain undecayed after 20 days?

 decay constant: $\lambda = \frac{A}{N} = \frac{60}{6.0 \times 10^7} \approx 1.0 \times 10^{-6} \text{ s}^{-1}$

number of remaining nuclei: $N = N_0 e^{-\lambda t} = 6.0 \times 10^7 \times e^{-1.0 \times 10^{-6} \times 20 \times 3600} \approx 1.1 \times 10^7$ $\square$

Example 23.6 $^{24}_{11}\text{Na}$, an isotope of sodium, has a decay constant of $1.28 \times 10^{-5} \text{ s}^{-1}$. Suppose a sample initially contains a mass of $9.0 \mu\text{g}$ of $^{24}_{11}\text{Na}$. Find (a) initial number of $^{24}_{11}\text{Na}$ nuclei, (b) initial activity of this sample, (c) the activity after 48 hours.

 initial number of nuclei: $N_0 = \frac{9.0 \mu\text{g}}{24\text{u}} = \frac{9.0 \times 10^{-9}}{24 \times 1.66 \times 10^{-27}} \approx 2.26 \times 10^{17}$

initial activity: $A_0 = \lambda N_0 = 1.28 \times 10^{-5} \times 2.26 \times 10^{17} \approx 2.89 \times 10^{12} \text{ Bq}$

activity after 20 hours: $A = A_0 e^{-\lambda t} = 2.89 \times 10^{12} \times e^{-1.28 \times 10^{-5} \times 48 \times 3600} \approx 3.17 \times 10^{11} \text{ Bq}$ $\square$

Question 23.11 A sample of polonium-205 has an activity of $6.7 \times 10^{15} \text{ Bq}$. The decay constant of polonium-205 is known to be $1.16 \times 10^{-4} \text{ s}^{-1}$. (a) Find the number of nuclei needed to give this activity. (b) Find the mass of polonium-205 in this sample. (c) Calculate the time needed for the activity reduce to 1% of its initial value.

Question 23.12 Technetium-99m is widely used as a radioactive tracer for medical diagnostic procedures. If some of this isotope, with an activity of 900 MBq was injected into a patient. The activity is found to reduce to 56.4 MBq after 24 hours. (a) What is the decay constant of technetium-99m? (b) How many technetium-99m nuclei are still left in the patient's body?

23.2.3 half-life

it is more convenient to define a time quantity to describe how fast radioactive nuclei decay
it is useful to define half-life of a radioactive sample

half-life ($t_{1/2}$) is the mean time taken for number of radioactive nuclei in the sample, or activity of the sample, to reduce to half of its initial value

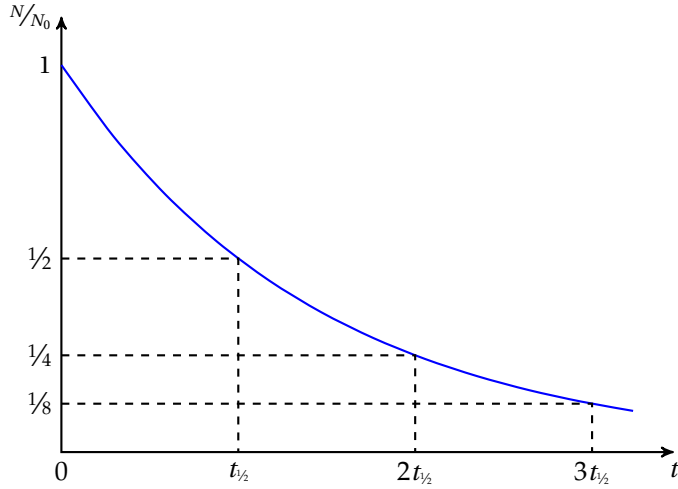
➤ half-life $t_{1/2}$ is closely related to decay constant λ

by definition, at $t = t_{1/2}$, $N = \frac{1}{2}N_0$, $A = \frac{1}{2}A_0 \Rightarrow e^{-\lambda t_{1/2}} = \frac{1}{2} \Rightarrow e^{\lambda t_{1/2}} = 2 \Rightarrow \lambda t_{1/2} = \ln 2$

➤ λ is a constant, so $t_{1/2}$ is also a constant over the lifetime of nuclear decay

half-life of a sample of isotope does not depend on initial number of nuclei or initial activity

Example 23.7 Radium-224 has a half-life of 3.63 days. (a) What is the activity from 6.0 mg of pure radium-224? (b) How many radium-224 nuclei have undergone decay after 10 days?



half-life of radioactive decay

decay constant: $\lambda = \frac{\ln 2}{t_{1/2}} = \frac{\ln 2}{3.63 \times 24 \times 3600} \approx 2.21 \times 10^{-6} \text{ s}^{-1}$

initial number of Ra-224 nuclei: $N_0 = \frac{6.0 \text{ mg}}{224 \text{ u}} = \frac{6.0 \times 10^{-6}}{224 \times 1.66 \times 10^{-27}} \approx 1.61 \times 10^{19}$

initial activity: $A_0 = \lambda N_0 = 2.21 \times 10^{-6} \times 1.61 \times 10^{19} \approx 3.57 \times 10^{13} \text{ Bq}$

number of undecayed nuclei: $N = N_0 e^{-\lambda t} = 1.61 \times 10^{19} \times e^{-2.21 \times 10^{-6} \times 10 \times 24 \times 3600} \approx 2.39 \times 10^{18}$

number of nuclei that have decayed: $\Delta N = N_0 - N = 1.61 \times 10^{19} - 2.39 \times 10^{18} \approx 1.37 \times 10^{19}$ $\square$

Example 23.8 Living trees contain a certain percentage of $^{14}_6\text{C}$, an radioactive isotope of carbon that has a half-life of 5570 years. A sample of dead wood is found to have an activity of 0.42 Bq, while an equal mass of living wood has an activity of 1.60 Bq. Find the age of the dead wood.

$A = A_0 e^{-\lambda t} = A_0 e^{-\frac{\ln 2}{t_{1/2}} t} \Rightarrow 0.42 = 1.60 e^{-\frac{\ln 2}{5570} t} \Rightarrow -\frac{\ln 2}{5570} t = \ln\left(\frac{0.42}{1.60}\right) \Rightarrow t \approx 10700 \text{ years}$ $\square$

Question 23.13 The number of uranium-238 nuclei in a rock sample is believed to have decreased from 3.50×10^{17} to 3.27×10^{17} in 480 million years. Estimate the half-life of uranium-238.

Question 23.14 Plutonium-238 is a powerful alpha emitter with a half-life of 87.7 years. One decay of plutonium-238 releases an energy of about $9.0 \times 10^{-13} \text{ J}$. A nuclear battery containing a sealed plutonium source is implanted into patient's body to power heart pacemakers. The battery has an initial activity of $6.0 \times 10^{10} \text{ Bq}$. (a) Calculate the initial power released by the source. (b) Find the mass of plutonium required to produce this power. (c) It is required that power output to the pacemaker is at least 60% of the initial power. Calculate the time, in years,

for which the battery provides sufficient power.

Question 23.15 Show that the variation of the number of undecayed nuclei with time t can be given by: $N(t) = N_0 \left(\frac{1}{2} \right)^{\frac{t}{t_{1/2}}}$, where N_0 is the initial number of nuclei.

23.2.4 measurement of radioactive decay

a Geiger–Müller tube, or a **GM tube**, is a device used to detect ionizing radiation

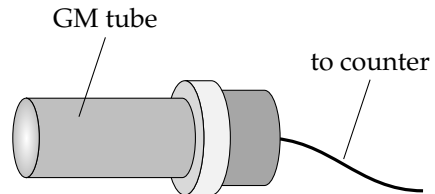
GM tube measures the number of α -particles, β -particles and γ -photons arriving per unit time

the number of decays recorded per unit time by

GM tube is called the **count rate** R

since GM tube only picks up emissions to one particular direction, but a radioactive source emits radiation in *all* directions, so count rate is a fraction of the activity of the sample

activity obeys exponential decay, so we would expect count rate to satisfy: $R = R_0 e^{-\lambda t}$



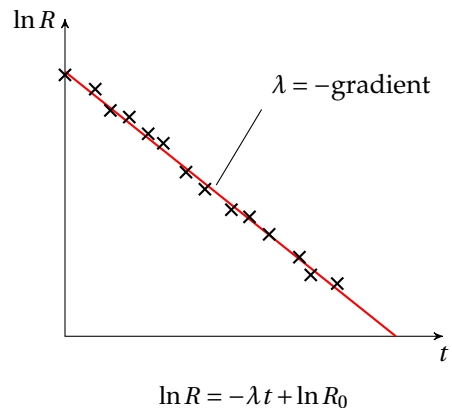
verifying the exponential decay law

to verify the exponential decay law, we first rearrange the equation as: $\ln R = -\lambda t + \ln R_0$

if a set of measurements of R at time t are obtained, a graph of $\ln R$ against t can be plotted

if trend curve shows a straight line, the relation $R = R_0 e^{-\lambda t}$ is then verified

decay constant λ is given by negative gradient of the best-fit line



issues with count rate

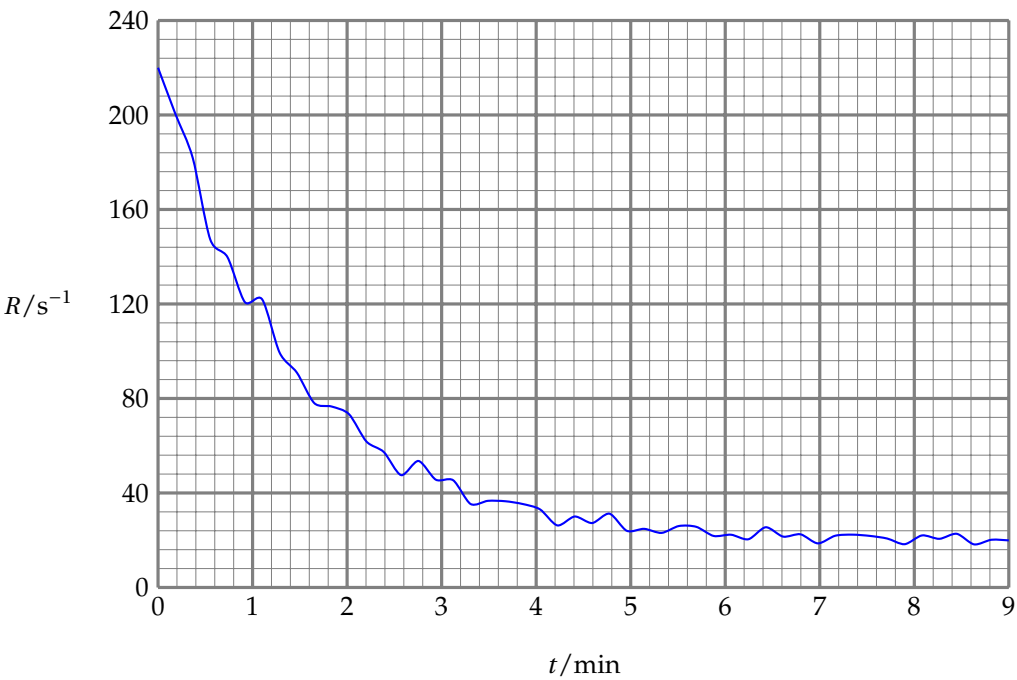
for now, we take for granted that the count rate measured is 100% accurate

but in practice, there might be a number of factors leading to measurement errors

- there is always *background radiation* from earth minerals, cosmic rays, food and water, etc.
background reading must be subtracted to give a corrected count rate
- α - and β -particles emitted could be absorbed by sample itself
- product nuclei could also be radioactive, contributing to additional counts
- GM tube has *dead time*, or *resolving time*

when a count is recorded, it takes GM tube a certain time to reset for the next count

Example 23.9 A GM tube is placed close to a source of radioactive isotope. The variation with time t of the measured count rate R is shown. Determine the half-life of this isotope.



there is evidence of *background radiation* as the count rate decreases and tends to a non-zero value of 20 s^{-1} , so we need subtract this number to obtain the true count rate of sample

there are also *fluctuations* in count rate due to *random* nature of decay processes, so we need to make several calculations and take average to obtain an accurate value for half-life

let's call true count rate R_T , and call measured count rate R

as $R_T = 200 \text{ s}^{-1} \rightarrow 100 \text{ s}^{-1}$, $R = 220 \text{ s}^{-1} \rightarrow 120 \text{ s}^{-1}$, so $t_{1/2} \approx 1.0 - 0.0 \approx 1.0 \text{ min}$

as $R_T = 160 \text{ s}^{-1} \rightarrow 80 \text{ s}^{-1}$, $R = 180 \text{ s}^{-1} \rightarrow 100 \text{ s}^{-1}$, so $t_{1/2} \approx 1.3 - 0.4 \approx 0.9 \text{ min}$

as $R_T = 120 \text{ s}^{-1} \rightarrow 60 \text{ s}^{-1}$, $R = 140 \text{ s}^{-1} \rightarrow 80 \text{ s}^{-1}$, so $t_{1/2} \approx 1.6 - 0.7 \approx 0.9 \text{ min}$

as $R_T = 80 \text{ s}^{-1} \rightarrow 40 \text{ s}^{-1}$, $R = 100 \text{ s}^{-1} \rightarrow 60 \text{ s}^{-1}$, so $t_{1/2} \approx 2.3 - 1.3 \approx 1.0 \text{ min}$

take average for these results, we find $t_{1/2} \approx 0.95 \text{ min}$

□